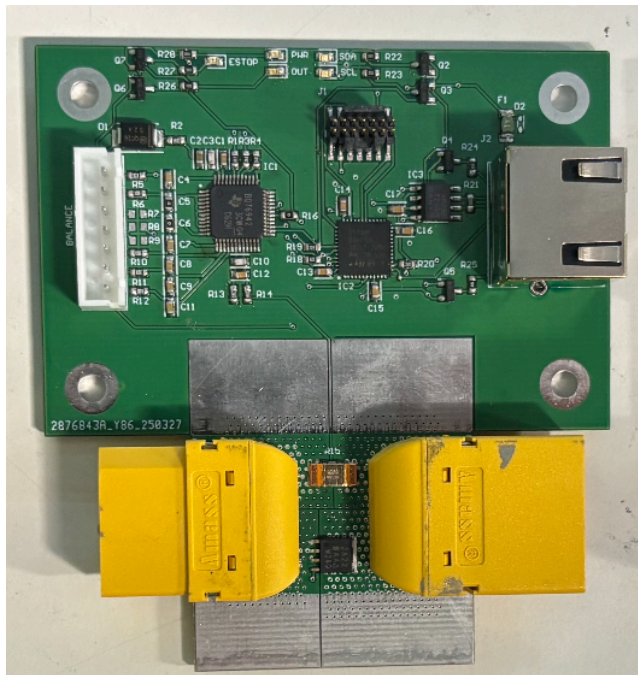


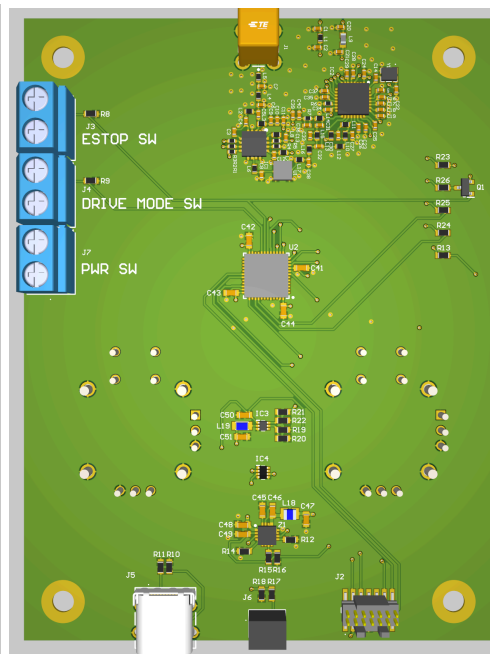
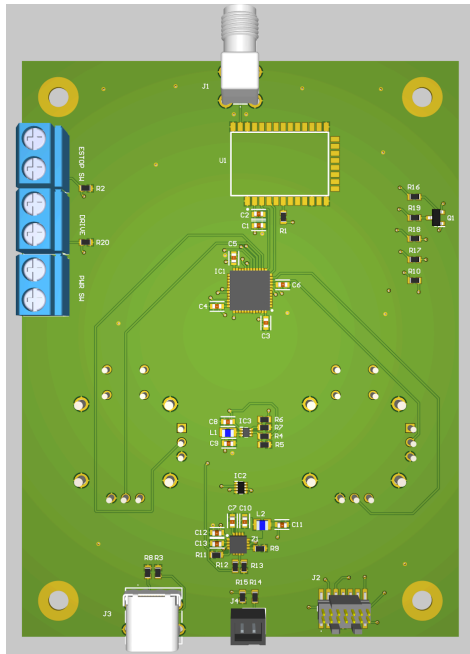
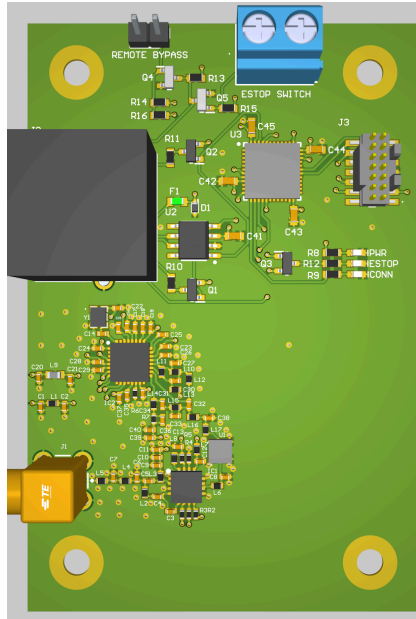
BMS



The BMS monitors battery health and performs battery shutoff. It protects against undervoltage, overcurrent, and cell imbalance. Last year, we implemented these functions with a microcontroller, leading to noisy measurements and unreliable performance. This year, we switched to the TI BQ76942, a battery monitor and protector for 3s-10s batteries. This chip provides many useful features. A built-in charge pump allows for a high side NFET, meaning communication with the rest of the system doesn't need to be isolated like the previous BMS. For our transistor, we use the Vishay SiJA22DP with an $R_{DS(on)}$ of 700 m Ω and current rating of over 200A. Additionally, the DFETOFF pin can be directly attached to the ESTOP line, reducing the chance buggy firmware would fail to turn off the FET in case of ESTOP. Finally, various quantities, including stack voltage, cell voltages, current, and FET state can be accessed over I2C from a microcontroller unit, which then sends these values over CAN to the rest of the system.

We have dealt with various issues with these boards. The goal is to have a single PCB that can be used with all batteries. On the first boards, the cell connections were not easily switchable between the 4s and 6s batteries. Additionally, sometimes the BMS would cycle when turning on. This was especially an issue with ESTOP, as we cannot have the risk of an indefinite cycle when turning off ESTOP mode. To overcome this, a revision fixed these cell connectors and added predischage functionality, allowing us to reduce the transient on turn on and hopefully fix the cycling issue. However, a further revision is needed to change the direction of the current sense resistor, and add a missing decoupling capacitor on the BMS IC.

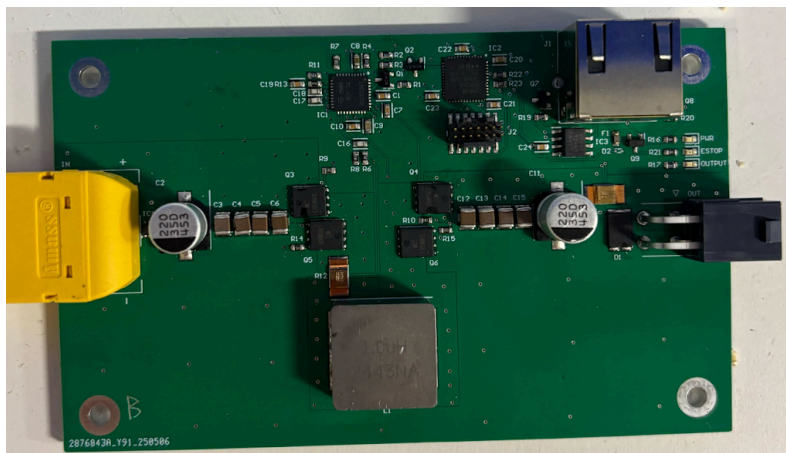
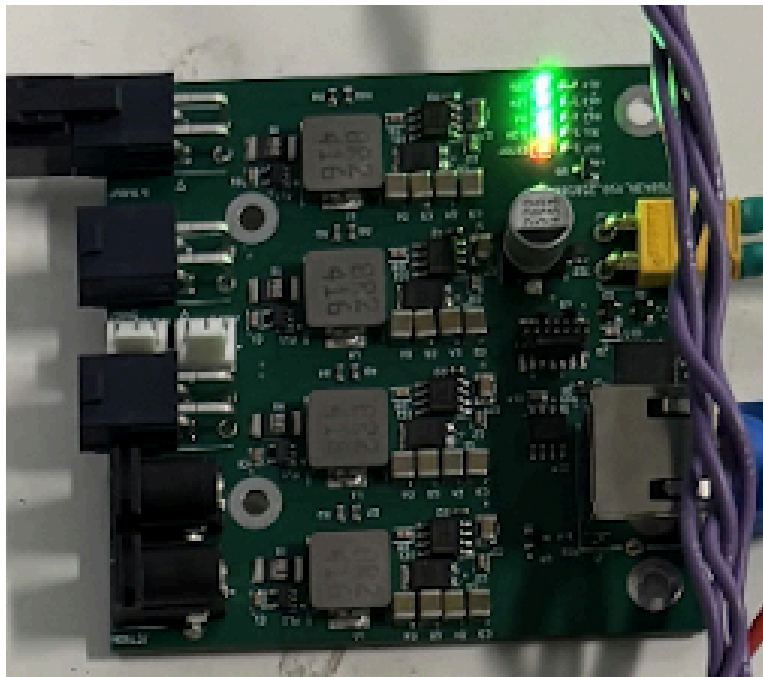
ESTOP



Last year, our ESTOP was implemented solely in the BMS, leaving the system susceptible to a failed short transistor. This year, we have redundant ESTOP mechanisms: a transistor on the BMS or a contactor can shut off the battery power. All PCBs receive the ESTOP signal, which is an active low signal (when the signal is a logic low, the ESTOP is activated). This means that the ESTOP will be triggered in the case of a broken ESTOP wire. Both the remote ESTOP and button on the boat trigger ESTOP by pulling the line low through a transistor, whose gate is pulled up as a failsafe measure. On the shore side, a custom controller sends ESTOP data, as well as manual drive data. If ESTOP is activated on the shore side, or connection is lost, ESTOP is triggered on the boat.

We have two methods of sending the ESTOP signal. Our primary method is to use Digi XBee XR 900 radios on the 915MHz ISM band, which utilize FHSS (Frequency Hopping Spread Spectrum) and feature built-in SAW filters, allowing for a stable connection in noisy RF environments. Our backup method is to use a circuit based on the TI CC1200 and CC1190, allowing us to customize the radio, as well for output powers up to 27 dBm.

Power

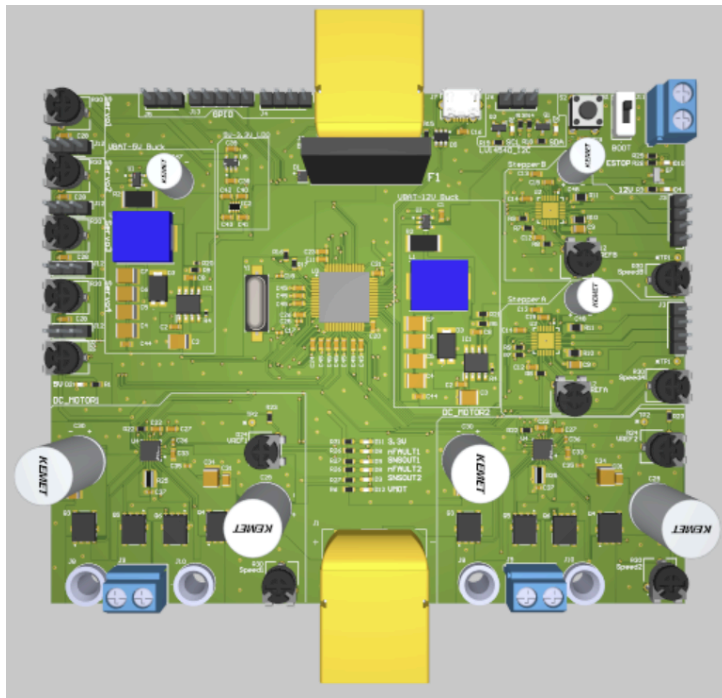


The boat is powered by three batteries: a 6-series (6s) battery for electronics, and two 4-series (4s) batteries for thrusters and high power mechanisms. The main power board contains buck converters that take the voltage from the electronics battery and convert it into 3.3V, 5V, 12V,

19V. The current of each of these rails is sent over CAN. The two mechanisms power boards feature a single DC-DC converter (using the TI LM251772 controller) that takes the voltage from a thruster battery and converts it to any voltage from 3.3V to 36V, with a max current of 25A. These converters feature adjustable voltage and current limits, as well as output current monitoring, all over the CAN bus.

The converters on the main power board use the TI LV14540, rated for 5A of continuous output. The current on each is measured with a shunt resistor, with the INA180A2IDBVR current sense amplifier. The mechanisms power contains the TI LM251772 4-Switch Buck-Boost Controller, which can be configured by an STM32. The STM32 can set the voltage and current limit, as well as read output current and fault status.

Mechanisms



This year, we built a mechanisms testing board, allowing for the mechanical team to test mechanisms easily without the overhead of the autonomy stack. This board can control servos, steppers, and DC motors, and can be configured using potentiometers or through a simple UI. This board connects to the mechanisms power boards, allowing ground testing to be performed with the same power supplies as the boat. From there, a translation to the boat mechanisms board is simple, only requiring firmware to interface with the CAN system.

The ground testing board features a STM32F103 microcontroller, two A4988 stepper motor drivers, two DRV8701 H-Bridge drivers powered by the mechanisms power board (allowing us to power these with the same exactly supply going on the boat), and four servo motor connectors. Buck converters provide 12V for the stepper motor supply, and 5V for the servo motor supply.